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PAD-JOB-0001 Pre-AGI Job Dossier

Purpose and scope

PAD-JOB-0001 is a public-good, continuously updatable "Pre-AGI Job Dossier" system: a rigorous, audit-friendly framework for measuring how *replaceable* occupations are **before** the availability of broadly autonomous, generally capable AGI systems (i.e., in a world dominated by today's and near-term frontier models plus integration tooling). It is explicitly designed to shift assessment away from layoff anecdotes and toward a task-based, mechanism-level accounting of how technology changes labour demand. This task-first framing is consistent with the "task approach" in labour economics, which treats tasks (not job titles) as the units that reallocate between labour and capital as technology changes.

A core design requirement is to distinguish **exposure** (tasks that could be affected) from **replacement** (net reduction in labour demand after accounting for adoption frictions, demand responses, and new-task creation). Global exposure work on generative AI repeatedly emphasises that exposure does not imply immediate full automation of entire occupations and often implies transformation and augmentation.

The dossier is meant to be usable by individuals, employers, unions, educators, and policymakers, but with publication safeguards that discourage weaponisation (e.g., using the index as a sole justification for layoffs, wage suppression, or discriminatory hiring). A governance-first posture is aligned with mainstream AI risk-management approaches that treat "governance" as a cross-cutting function (govern → map → measure → manage) rather than as an afterthought.

Because the user did not specify a target set, this report proposes sensible defaults: a first-release "core coverage" across 12 high-priority sectors and 24 representative occupations (defined later). These defaults prioritise (i) high employment share, (ii) high task digitisation, (iii) high expected generative-AI exposure, and (iv) roles where labour-market signalling and policy relevance are high.

Definitions and conceptual model

The dossier's definitions must be stable, crosswalkable across countries, and operationalisable in data. The most robust starting point for international comparability is the "job/occupation" distinction used in ISCO-08: a **job** is "a set of tasks and duties performed… by one person," while an **occupation** is a set of jobs with similar main tasks and duties.

To connect international definitions to task-level measurement, the dossier uses occupational descriptor systems that explicitly enumerate tasks, work activities, and work context. The database is a practical anchor because it provides a rich set of variables describing work and worker characteristics and supports occupation-level "details" reports containing descriptor ratings across tasks, work activities, work context, skills, knowledge, and abilities.

Job versus task (and why this matters)

A **task** is the smallest unit of work that can be assigned, performed, and evaluated with meaningful completeness (e.g., "enter invoice data into ERP," "draft a first-pass customer email," "triage a patient," “pick items from bin locations”). Jobs are best treated as **weighted bundles of tasks**; the weights represent time share, importance, frequency, or criticality. The task approach literature cautions that technological change can (a) automate subsets of tasks, (b) re-bundle tasks across occupations, and (c) create new tasks, so job titles are not reliable “atoms.”

Replaceability, exposure, adoption, displacement, transformation

PAD-JOB-0001 separates five related constructs:

- **Exposure**: the share of tasks that are *technically* within scope of a capability set (e.g., generative AI applied to language-heavy tasks). Exposure frameworks for LLMs estimate occupation-task alignment without asserting a specific adoption timeline.
- **Feasibility/reliability**: whether systems can perform those tasks to required quality under real operating constraints (edge cases, security, low tolerance for errors).
- **Adoption**: whether organisations actually deploy systems at scale, conditional on integration costs, governance requirements, and the economic payoff. Enterprise adoption surveys show many organisations report AI use, but scaling and workflow redesign remain major bottlenecks, reinforcing the need to model adoption rather than assuming it.
- **Displacement (net labour demand reduction)**: the extent to which automation substitutes for labour *in equilibrium*, accounting for demand growth, price effects, and new tasks. Task-based macro frameworks explicitly treat automation as a displacement channel and new tasks as a reinstatement channel.
- **Transformation**: changes in task mix, skill gradients, supervision structures, and role boundaries—even when headcount does not fall immediately. Empirical evidence from generative AI in customer support shows productivity gains with heterogeneous worker impacts (e.g., larger gains for newer workers), illustrating transformation and learning dynamics rather than instantaneous role deletion.

Automation ratio (task automation ratio) and related terms

To make “replaceability” measurable and updateable, PAD-JOB-0001 defines:

- **Task Automation Ratio (TAR)**: the expected share of an occupation’s current task-time (or task-importance weight) that could be automated to acceptable quality *within a defined scenario and time horizon*, ignoring adoption barriers. This is close to “exposure,” but should be explicitly conditioned on capability assumptions.
- **Effective Automation Ratio (EAR)**: TAR adjusted for adoption frictions (integration, compliance, trust), supervision bandwidth, and cost curves. This bridges “potential” to “plausible deployment.”
- **Job Replaceability Index (JRI)**: a 0–100 composite score intended to approximate *conditional* replaceability (higher = more replaceable), reporting uncertainty intervals and sensitivity to assumptions, consistent with composite-indicator best practice.
- **Human Comparative Edge Density (HCED)**: the share of a task bundle where humans retain a persistent comparative advantage because of requirements such as relationship trust, legal/ethical legitimacy, embodied dexterity, or high-stakes responsibility. Trust frictions are substantive: research on “algorithm aversion” shows people can avoid algorithmic recommendations after seeing errors even when algorithms outperform humans, making trust a real constraint in adoption and scope.

Methodology

PAD-JOB-0001 should be built as an *evidence-led pipeline* that transforms occupational/task descriptors into scenario-conditioned replaceability measures with explicit uncertainty. This section specifies a methodology that is (i) implementable with existing datasets, (ii) robust under scrutiny, and (iii) designed for continuous updating.

Task decomposition

For each occupation, the dossier constructs a task inventory with weights:

1. **Canonical occupation definition and code**: store crosswalks (e.g., SOC ↔ ISCO ↔ local classification like SSOC where relevant). The 2018 SOC system is the US federal standard for classifying jobs for statistical purposes, enabling linkage to many labour datasets.
2. **Task list**: ingest tasks and descriptor ratings from O*NET (tasks, detailed work activities, work context). O*NET task/descriptor outputs are explicitly designed for export to structured formats and provide the substrate for systematic scoring.
3. **Task weights**: estimate time share/importance by combining O*NET importance ratings with role-specific expert judgement and (where available) employer process maps or time-and-motion studies. The system must allow "contestable edits" where domain experts revise task weights with justification to reduce description laundering.

Capability mapping

Map each task to capability primitives and performance requirements:

- **LLM/GenAI task mapping**: use published exposure rubrics that classify which tasks are affected by LLMs and which are accelerated by complementary tooling. For example, task-based exposure frameworks estimate the task shares impacted by LLMs and stress that they do not predict a specific adoption timeline.
- **Global exposure calibration**: incorporate the refined global index approach that combines task-level data, expert input, and AI model predictions, explicitly emphasising transformation and augmentation rather than total automation for many roles.

Capability mapping must include *non-text modalities* (vision, speech, robotics, sensors) and explicit **error tolerance** thresholds so that "can do it sometimes" is not scored the same as "can do it reliably" in production settings. This aligns with risk-management framings that require context-specific evaluation rather than abstract capability claims.

Cost curves and unit economics of automation

Replaceability is scenario-dependent because adoption depends on whether automation is economically rational:

- **Wage baseline**: use official wage and employment estimates such as OEWS from the , noting scope limits (wage and salary workers in nonfarm establishments; excludes self-employed in standard estimates). These data provide a defendable wage denominator for "wage-to-automation-cost" comparisons.
- **Adoption reality check**: incorporate enterprise adoption evidence and scaling constraints. For example, McKinsey's global surveys report high rates of AI use in at least one business function,

while simultaneously emphasising that many organisations have yet to scale, underscoring the need to separately model “pilotable” versus “scalable” automation.
- **Sectoral labour-market context**: use labour-market macro context and scenario pathways (e.g., growth, labour income share pressures) from ILO macro updates to avoid interpreting technical feasibility as inevitable displacement.

Supervision bandwidth

Even when tasks can be automated, they may require substantial human oversight:

- **Operational definition**: supervision bandwidth is the ratio of “human oversight hours” required per unit of automated output to meet quality/safety constraints. It is a *capacity constraint* that can bottleneck scaling.
- **Regulatory alignment**: high-risk domains often require human oversight responsibilities and competence/training for oversight roles. The EU AI Act provides explicit language on human oversight mechanisms and the need for assigned humans with competence, training and authority.
- **Risk governance practice**: NIST AI RMF structures risk management into govern/map/measure/manage, consistent with continuously monitoring deployed system performance rather than scoring replaceability as a one-off.

Supervision bandwidth is also where “job deletion” narratives often fail: roles can become “AI-supervised work” rather than disappear, shifting skill composition and creating new coordination tasks.

Regulatory friction and trust premium

- **Regulatory friction**: model as (i) licensing requirements, (ii) compliance costs, (iii) liability exposure, and (iv) mandatory documentation/testing. The EU AI Act’s conformity and oversight framing is a concrete template for why some deployments will be slower and more expensive in regulated environments.
- **Trust premium**: the “extra” human involvement required for legitimacy, acceptance, or relationship needs. Behavioural evidence shows that people can be markedly averse to algorithms after observing errors (“algorithm aversion”), which can limit adoption even when performance is strong.

Data sources and evidence spine

PAD-JOB-0001 should prioritise primary/official sources, then peer-reviewed or working-paper research, then reputable industry measurement. The goal is not to collect “more sources,” but to maintain an updateable “evidence spine” where each metric has (a) an upstream dataset, (b) a refresh cadence, (c) versioning and provenance.

Recommended primary and official sources

The table below is a default “evidence spine” that is both defensible and practically downloadable.

Evidence spine component	What it provides	Why it matters for PAD-JOB-0001	Suggested refresh
O*NET database + occupation reports	Task lists, work activities, work context, skill/ability descriptors; exportable structured data	Enables task decomposition and feature extraction for scoring	With each O*NET release
OEWS (BLS)	Employment and wage estimates by occupation (with scope limits)	Anchor for wage baselines and sector/ occupation weighting by employment	Annual / with BLS releases
ISCO-08 definitions and structure	Internationally comparable definition of job/occupation	Enforces conceptual clarity and cross-country comparability	Stable; revised infrequently
SOC 2018 manual	Official US occupational classification	Provides standard codes for linking US datasets (BLS, O*NET)	With SOC revisions
ILOSTAT employment by occupation	Cross-country occupational employment; includes modelled estimates	Supports global/region weighting and distributional analysis	As ILOSTAT updates
Eurostat LFS (ISCO-coded employment)	Europe employment by detailed occupation	Enables region-specific sector/occupation weighting and validation	Quarterly/ annual (Eurostat)
NIST AI RMF 1.0	Governance and risk-management structure for AI	Templates the dossier's governance, monitoring, and audit loop	Stable; profile addenda evolve
EU AI Act (Reg. 2024/1689)	Compliance/oversight requirements for high-risk AI	Anchors regulatory friction and oversight constraints	Track amendments/ guidance

If a Singapore-localised edition is desired, a practical starting point is the Singapore Standard Occupational Classification (SSOC), which adopts the ISCO framework, plus the Ministry of Manpower's labour force statistical releases.

Core research references to operationalise (academic / institutional)

The following are "load-bearing" because they define the conceptual separation between tasks, exposure, adoption, and net impacts:

- Task-based labour market theory and measurement: on the task approach; and with on automation displacement versus new-task reinstatement.
- LLM exposure measurement: et al. propose a rubric for assessing occupation-task exposure to LLMs and stress the difference between model capability and timelines/adoption.
- Empirical workplace impacts: , , and study a deployed GenAI assistant in customer support and find measurable productivity gains with heterogeneous effects across experience levels.

- Composite indicator construction and robustness: the OECD handbook provides widely cited guidance on constructing, weighting, and testing composite indicators (including sensitivity/uncertainty).
- Trust and adoption constraints: et al. document algorithm aversion and show that even slight perceived loss of control can reduce adoption, motivating explicit modelling of trust premiums rather than assuming frictionless substitution.

Industry and policy context sources can serve as scenario inputs: employer surveys on skills/jobs and planning horizons; AI Preparedness Index for national capacity context; and AI Index for trend baselines on the economy and labour demand signals.

Recommended citation style for the dossier

A public-good dossier benefits from a hybrid of (i) academic readability and (ii) data provenance. A practical standard is:

- **Author–date (Chicago or APA-like) for papers**, with DOI/arXiv/NBER identifiers where available.
- **Dataset citation with version + access date + publisher**, e.g., "BLS OEWS (May 2024 estimates), accessed YYYY-MM-DD; O*NET database release X.Y; ILOSTAT download (indicator ID), accessed YYYY-MM-DD."
- **Evidence ledger discipline**: every metric field should record its upstream source, retrieval date, version hash, and transformation steps (data statement).

This approach is consistent with the OECD handbook's emphasis on transparency in choices (selection, normalisation, weighting) because composite systems are especially vulnerable to false precision when provenance is unclear.

Metrics, index formula, and validation plan

PAD-JOB-0001 should expose both *task-level* measures (transparent and contestable) and *occupation-level* composites (useful for navigation and comparison). The key is that the composite must remain decomposable: users must be able to see what drives a score.

High-priority sectors and representative occupations

The table below proposes 12 first-release sectors and 24 representative occupations (two per sector). These are defaults, not an exhaustive taxonomy.

Priority sector	Representative occupations (default set)	Rationale for inclusion
Clerical & administrative	Administrative Assistant; Data Entry Clerk	High digitisation and high GenAI exposure potential via language/document tasks
Customer operations (BPO/contact centres)	Customer Service Representative; Call Centre Supervisor	Strong evidence base on GenAI workplace impacts and augmentation dynamics
Finance & accounting ops	Accounts Payable/Receivable Clerk; Payroll Specialist	Structured documents + compliance constraints; clear wage/cost modelling using official wage data

Priority sector	Representative occupations (default set)	Rationale for inclusion
Finance (analysis)	Financial Analyst; Budget Analyst (or FP&A)	Mix of automation and judgement; useful for modelling “human legitimacy” and trust constraints
Legal & compliance	Paralegal; Compliance Officer	High text exposure + high regulatory liability; tests “regulatory friction” mechanism
Education & learning	Primary School Teacher; Instructional Designer	High relational/legitimacy components; tests HCED and trust premium
Healthcare (clinical delivery)	Registered Nurse; Radiologic Technologist	Safety-critical + regulated + trust; showcases adoption friction and oversight needs
Healthcare (administration)	Medical Coder; Medical Records Specialist	High digitisation and documentation; strong automation-ratio modelling
Software & IT	Software Developer; IT Support Specialist	Core “copilotisation” sector; high transformation variance
Media & marketing	Copywriter; Graphic Designer	High near-term tool impact; strong scenario sensitivity and adoption variance
Manufacturing	CNC Machine Operator; Quality Inspector	Tests multimodal/physical constraints; separates digital vs embodied work
Logistics, transport, hospitality, care (service economy edge cases)	Warehouse Picker/ Packer; Delivery Driver (last-mile)	High employment share globally; clarifies “physical presence requirement” and supervision bandwidth limits

Core metric set

The dossier should publish a small set of “headline” metrics, plus a deeper decomposed layer model. A recommended minimal headline set is below.

Metric	Definition	Range	Direction (higher = more replaceable?)	Primary anchors
TAR (Task Automation Ratio)	Expected share of task bundle automatable to acceptable quality under scenario/horizon	0–1	Yes	LLM exposure frameworks + task lists

Metric	Definition	Range	Direction (higher = more replaceable?)	Primary anchors
HCED (Human Comparative Edge Density)	Share of task weight requiring persistent human comparative advantage (trust/legitimacy/embodiment)	0–1	No (a "moat")	Trust and oversight literature; regulatory constraints
AFS (Adoption Friction Score)	Composite of integration complexity, compliance cost, vendor risk, downtime tolerance	0–1	No (a "moat")	NIST AI RMF; regulatory regimes
SB (Supervision bandwidth)	Oversight hours required per automated output unit to maintain quality/safety	≥0	No (higher SB limits scaling)	Human oversight requirements + risk management
JRI (Job Replaceability Index)	Composite 0–100 replaceability score, reported with uncertainty	0–100	Yes	OECD composite indicator guidance
CI (Confidence interval band)	JRI percentiles (p10/p50/p90) from Monte Carlo over uncertain inputs	—	—	OECD robustness/sensitivity

Index construction and formulas

A practical, auditable structure is:

1. **Task-level scoring**
For occupation j with tasks $t \in T_j$, define a task weight w_{jt} (summing to 1).
Define $P_{auto}(t \mid s)$: probability a task can be automated under scenario s (capability + reliability constraints).

Task Automation Ratio:

$$TAR_j(s) = \sum_{t \in T_j} w_{jt} \cdot \mathbb{E}[P_{auto}(t \mid s)]$$

1. **Adoption and constraint adjustment**
Define adoption probability $A_j(s)$, derived from integration cost, compliance, supervision bandwidth constraints, and trust premium. This step operationalises the fact that capability alignment is not the same as real deployment.

Effective Automation Ratio (one defensible form):

$$EAR_j(s) = TAR_j(s) \cdot A_j(s)$$

1. **Composite JRI (0–100)**
Use a transparent composite with named components and "moat" inversion (so frictions reduce

replaceability). The OECD handbook recommends explicit choices for normalisation, weighting, and uncertainty/sensitivity testing because rankings are especially sensitive to methodological choices.

Baseline form (layered, with subdimensions beneath each layer):

$$JRI_j(s) = 100 \cdot \sum_{k=1}^{K} \alpha_k(s) \cdot \frac{score_{jk}(s)}{100}$$

where $score_{jk}$ are normalised 0–100 layer (or subdimension) scores and α_k are scenario-specific weights (summing to 1).

1. **Uncertainty intervals (confidence bands)**
   Represent each uncertain input as a distribution (triangular min/mode/max is often usable for expert-elicited ranges), then run Monte Carlo to compute $\mathbb{E}[JRI]$ plus percentiles (p10/p50/p90). This directly supports "confidence grading" and avoids false precision.

2. **Sector indices (weighting)**
   To produce sector-level dashboards, use employment-weighted aggregation:

   $$JRI_{sector}(s) = \sum_{j \in sector} share_{employment,j} \cdot JRI_j(s)$$

   Employment shares should come from official labour stats (BLS/ILOSTAT/Eurostat or local systems like SSOC-based tables).

Demonstration charts and sensitivity analysis

The two figures below are **illustrative demos** generated with synthetic placeholder data to show what PAD-JOB-0001 outputs *should look like* once task scoring is implemented end-to-end (they are not asserted estimates of the occupations listed).

Illustrative distribution of expected JRI (demo)

Illustrative sensitivity: rank range across scenario weight-sets (demo)

A downloadable demo table (24 occupations) is here: Download CSV

Validation plan

A pre-AGI replaceability dossier must be validated on *three axes*: measurement validity, predictive usefulness, and governance integrity.

Measurement validity (are we measuring what we claim?) should combine:

- **Case studies with known deployments**: start with domains where quasi-experimental evidence exists (e.g., customer support GenAI assistants). Empirical results on productivity and heterogeneous worker impacts provide a concrete anchor for calibrating transformation versus replacement assumptions.

- **Cross-check against global exposure indices**: compare TAR distributions and “affected task share” against ILO exposure indices and LLM exposure frameworks to ensure the dataset is not systematically under/over-estimating exposure in language-intensive tasks.
- **Crosswalk sanity checks**: ensure SOC↔ISCO mapping does not silently shift task bundles; use official manuals/definitions as the invariants.

Predictive usefulness (does it help decisions?) should use:

- **Expert interviews with structured elicitation** (not “vibes”): require task-level edits, min/mode/max ranges, and explicit rationales. Composite indicator guidance emphasises transparency in weighting and robustness checks to prevent spurious certainty.
- **Longitudinal updates and back-testing**: every release should store prior predictions, then compare against observed changes in task composition, tooling adoption, or occupational demand signals. Macro and institutional sources (OECD labour studies; AI Index labour demand indicators) can provide context for interpreting observed labour outcomes.

Governance integrity (can it be gamed or misused?) should include:

- **Red-team misuse tests**: prompts and protocols designed to detect attempts to use the index for discriminatory or coercive ends (e.g., “rank which jobs to fire first”). Risk management frameworks emphasise governance and monitoring as ongoing functions rather than one-off compliance.
- **Appeals and contestability**: stakeholders must be able to contest task lists, weights, and assumptions; otherwise, the system is structurally vulnerable to description laundering and lobbying—classic failure modes for high-stakes composite indicators.

Deliverables, implementation roadmap, risks, and recommended next steps

Deliverables

PAD-JOB-0001 should ship as a “dossier stack” rather than a single PDF:

1. **Executive summary (public, plain language)**: definitions, what the index is and is not, headline sector patterns, uncertainty warnings. This should mirror ILO-style clarity on “exposure ≠ automation of entire occupations.”
2. **Technical whitepaper**: full metric specs, scoring rubrics, data provenance, uncertainty math, sensitivity tests, validation results, and governance procedures. Composite indicator best practice requires explicit documentation of normalisation/weighting and robustness analyses.
3. **Visual heatmaps and scenario dashboards**: sector-by-occupation matrices with TAR/HCED/AFS/JRI and scenario deltas. Scenario design can be informed by employer survey horizons and macro labour context (WEF and ILO).
4. **Transition playbooks (role-specific)**: “augmentation patterns,” supervision and workflow redesign, skills portfolios, and credential pathways—grounded in observed transformation dynamics (e.g., stronger impacts for novices in at least one well-studied workplace deployment).
5. **API + dataset release**: versioned occupation entries, task inventories, metric values, and evidence ledgers.

Sample dossier entry template (occupation-level)

This template is meant to be continuously updateable and audit-friendly.

Field	Example content (template)
Occupation name	“Customer Service Representative”
Classification codes	SOC-2018: ####-####; ISCO-08: ####; (optional) SSOC: ####
Scope note	“Wage & salary roles; excludes self-employed variants unless separately modelled”
Task inventory	List of tasks with weights; source tags (O*NET task ID / expert edit / employer process map)
Capability map	Task→capability primitive mapping; failure modes/edge cases; required modalities
TAR / HCED / AFS / SB	Numeric values + ranges; distributions (min/mode/max)
JRI	Expected value + (p10, p50, p90); confidence grade; scenario deltas
Drivers	Top contributors, with links to task items and evidence
Transformation vector	Probabilities across: Replace / Augment / Unbundle / Elevate / Create (must sum to 1)
Governance	Changelog; reviewer list and COI statements; appeal instructions; model/data versions
Evidence ledger	For each key claim: source, year, retrieval date, and what it supports

API / data schema (minimum viable)

A minimal “public dataset” can be structured as two tables/objects: `occupation` and `task`.

Occupation object (JSON shape)
- `occupation_id` (stable UUID)
- `occupation_name`
- `codes` {`isco08`, `soc2018`, `local_optional`}
- `sector`
- `metrics` {`tar`, `hced`, `afs`, `sb`, `jri_expected`, `jri_p10`, `jri_p50`, `jri_p90`, `confidence_grade`}
- `scenario_scores` (map of scenario→JRI distribution summary)
- `drivers` (ranked list of dimension contributions)
- `transformation_vector` {`replace`, `augment`, `unbundle`, `elevate`, `create`}
- `provenance` {`data_versions`, `model_versions`, `reviewers`, `last_updated`, `next_review_due`}

Task object
- `task_id`
- `occupation_id` (FK)
- `task_text`
- `task_weight`
- `capability_tags` (multi-label)
- `p_auto_distribution` (min/mode/max or parametric family)

- `human_edge_tags` (trust/legitimacy/embodiment/etc.)
- `evidence_refs` (IDs into an evidence ledger)

This approach makes the index decomposable: the JRI can always be reconstructed from task-level assumptions and documented weights—an explicit transparency requirement for composite indicators used in policy contexts.

Implementation roadmap

A no-budget-constraint roadmap should still be staged to reduce model risk.

Phase framing (build the spine first, then scale coverage):

- **Foundation**: establish crosswalks, data ingestion, and evidence ledger discipline (O*NET + SOC + ISCO + labour stats).
- **Scoring engine**: implement task decomposition, capability mapping, and Monte Carlo uncertainty; bake in sensitivity tests (weight perturbations, leave-one-dimension-out), consistent with OECD composite indicator guidance.
- **Validation loop**: select 3–6 deep case-study occupations (including at least one with strong empirical evidence like customer support) to calibrate transformation assumptions.
- **Governance operations**: implement the risk governance workflow and public change logs; use NIST AI RMF as the structural template for operate-and-monitor rather than publish-and-forget.
- **Scale-out**: expand from 24 to 200+ occupations with triage rules: expand by employment share (BLS/ILOSTAT/Eurostat), policy salience, and stakeholder requests.

Update cadence (recommended): - **Monthly**: capability mapping updates (model/tool progress signals), with explicit versioning. Trend monitoring can be informed by ongoing AI Index releases and enterprise adoption reporting.
- **Quarterly**: scenario refresh (macro conditions, regulation changes, adoption barriers).
- **Annual**: labour stats refresh (wages, employment weights), aligned to official release schedules.

Open licensing (public-good posture): - Publish text under Creative Commons (e.g., CC BY 4.0 or CC BY-SA 4.0), code under a permissive licence (e.g., Apache-2.0), and datasets under an open data licence consistent with upstream constraints. This is an implementation choice, but it supports broad reuse while keeping provenance intact.

Risks and limitations (with mitigation)

Risk: confusing exposure with replacement. Exposure indices often show large task shares affected by GenAI, but both research and global institutional work warn that exposure ≠ immediate automation of entire occupations. Mitigation: publish TAR separately from JRI and require scenario conditioning and uncertainty in every public view.

Risk: false precision and unstable rankings. Composite indicators are sensitive to normalisation and weighting; without sensitivity/uncertainty analysis, rankings can be brittle and misleading. Mitigation: mandatory Monte Carlo uncertainty bands, scenario weight-sets, and robustness reporting.

Risk: Goodharting, lobbying, and "job description laundering." High-stakes indices attract gaming. Mitigation: contestable task inventories with audit logs, multi-stakeholder review, and clear change-control. Composite indicator guidance explicitly foregrounds transparency and robustness testing to limit manipulation via methodological choices.

Risk: regulatory and trust shocks. Regulation and trust can abruptly shift adoption pathways. Mitigation: treat regulation and trust as explicit subdimensions; map high-risk domains and oversight requirements using primary legal/risk frameworks.

Risk: misuse against workers. Even accurate indices can be weaponised. Mitigation: publish with "safe use" policies, aggregation-first reporting, and refusal guidelines; operationalise governance as ongoing monitoring and accountability.

Recommended next steps

1. **Lock the v1 conceptual contract**: finalise definitions (job/task/replaceability), the separation of TAR vs JRI, and scenario horizons, grounded in ISCO definitions and task-based labour theory.
2. **Build the evidence spine and crosswalks**: implement O*NET + SOC + labour stats ingestion, plus an evidence ledger format that is versioned and reproducible.
3. **Pilot 6 deep dossiers + 18 light dossiers**: use a small number of deep case-study occupations (including customer support) to calibrate transformation dynamics with empirical anchors.
4. **Ship the first public dashboard with uncertainty**: publish p10/p50/p90 bands and sensitivity notes from day one to avoid false determinism, consistent with composite-indicator best practice.
5. **Stand up governance and a quarterly review cycle**: implement a govern→map→measure→manage loop (NIST AI RMF) and track regulatory changes that materially affect adoption friction.
